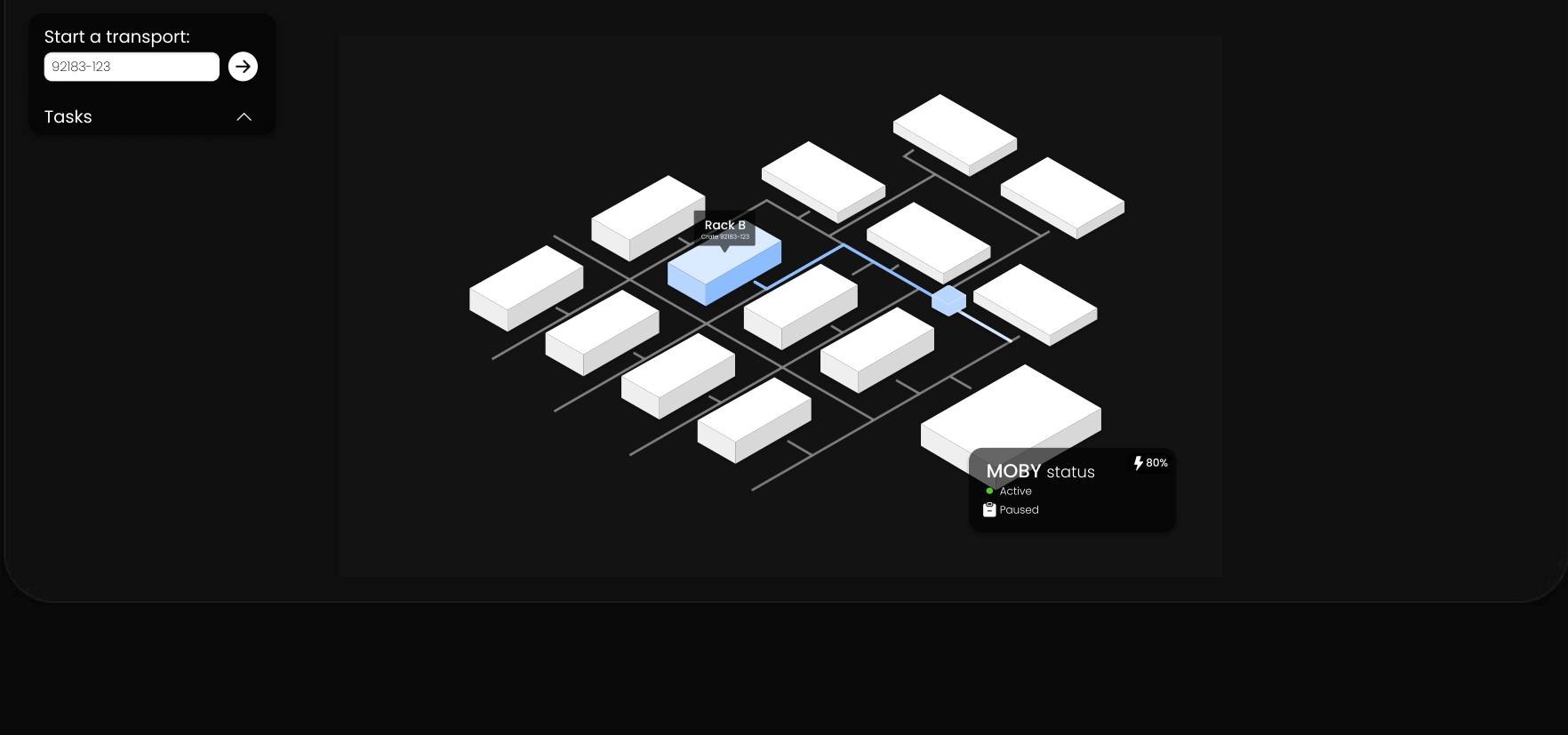


Building a scalable interface foundation for robotics.

A complete GUI redesign and scalable component system built for Neura Robotics — bringing visual consistency, operator clarity, and long-term maintainability to a rapidly evolving industrial platform.

- UX/UI Design
- Design System
- Components
- Workflows



01 — THE PROBLEM

A powerful system, but visually disorganized.

The Neura Robotics control GUI grew organically over several years of rapid feature additions. Without a design system to anchor decisions, each team made independent choices — resulting in visual inconsistencies, duplicated components, and an operator experience that felt fragmented and difficult to learn.

+40 Screens
The interface had grown to over 40 screens with no unified structure, making it hard to navigate and maintain.

Inconsistencies
Buttons, inputs, and panels differed across screens — no shared tokens, no single source of truth for visual language.

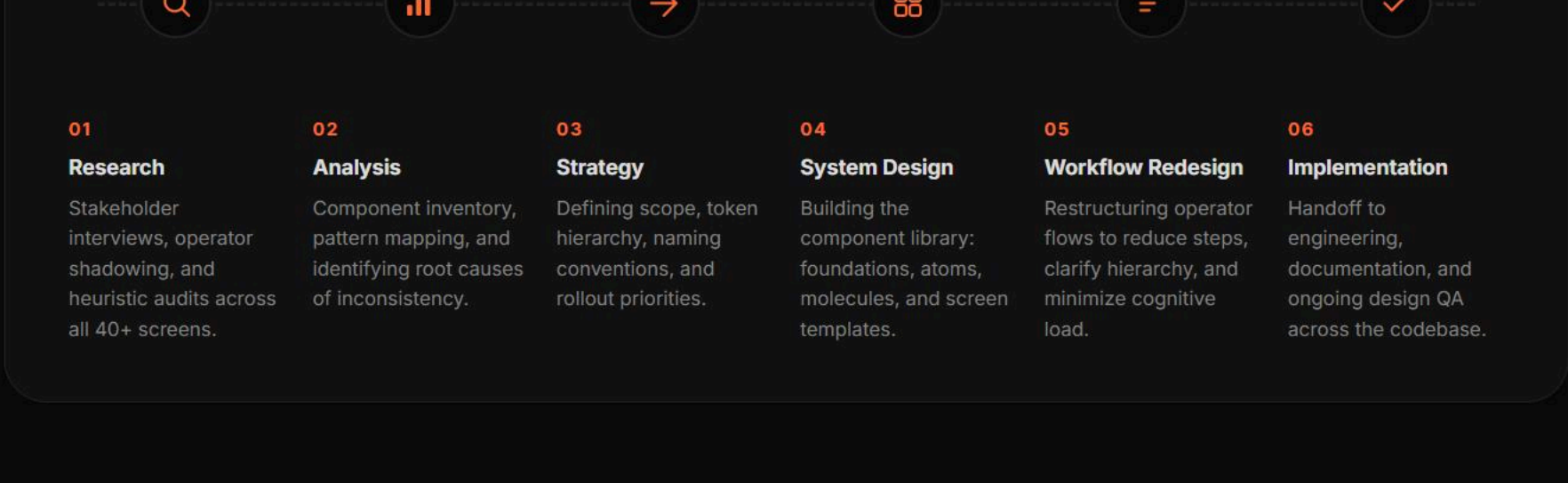
Duplication
Identical UI patterns were rebuilt from scratch for each feature, accumulating technical and design debt over time.

Costly Maintenance
Every design change required touching multiple disconnected files, multiplying effort and increasing the chance of regressions.

Unclear UX
Operators struggled to locate critical controls and understand feedback states, increasing errors during high-pressure tasks.

02 — PROCESS OVERVIEW

A process focused on understanding, structuring, and scaling.



03 — DESIGN SYSTEM

A modular, consistent system.

Foundations
Color tokens, typography scale, spacing units, grid definitions, and motion values — the raw material for every decision.

Components
Buttons, inputs, badges, modals, tooltips, and data displays — each with documented states and behavior variants.

Patterns
Reusable layout templates and interaction patterns composed from components: forms, dashboards, data tables, alert flows.

Screens
Full operator interfaces assembled from patterns — production-ready, version-controlled, and reviewed by engineering QA.

COLOR TOKENS

TYPE SCALE
Aa Display / 800
Aa Heading / 700
Aa Body / 400
Aa Label / 600

COMPONENTS

Primary Secondary Ghost

Joint velocity (rad/s)

Active Idle Error Standby

SCREEN PREVIEW

Joint Temp: 42°C Payload: 7.2 kg

Uptime: 99.1% Errors: 2

04 — WORKFLOW REDESIGN

Simpler, clearer, more efficient flows.

Before
ARM CALIBRATION FLOW
Home → Settings → Hardware → Calibrate → Joint Config → Confirm → Reboot
7 steps • 3 context switches
Issues

- Calibration buried 4 levels deep
- No inline feedback on joint state
- Settings split across 3 panels
- Required reboot on every change
- No undo or preview mode

After
ARM CALIBRATION FLOW
Home → Quick Calibrate → Apply
3 steps • inline • no reboot
Improvements

- Calibration accessible from main toolbar
- Real-time joint state feedback inline
- Unified settings panel with search
- Live apply without full reboot
- Undo + preview mode for all changes

OPTIMIZED FLOWS
12
operator task paths redesigned

STEPS REDUCED
-60%
avg. interaction depth per task

COGNITIVE LOAD

fewer decisions at critical moments

05 — IMPACT

A solid foundation to scale the future.

3x
Faster Development
Engineers ship new screens in a third of the time using the shared component library vs. building from scratch.

100%
Consistency
Every screen now draws from the same token set. Visual drift across product areas is eliminated.

240+
Reusable Components
Over 240 production-ready components documented with usage guidelines, variants, and accessibility notes.

Scalability
New product lines and hardware generations can adopt the system without redesigning from scratch.

06 — CLOSING

This project proved that design systems in industrial contexts aren't just a nice-to-have — they're the infrastructure that allows complex, safety-critical software to evolve without collapsing under its own weight. By investing in a structured foundation early, the Neura Robotics team now ships with confidence, clarity, and consistency across every screen operators depend on.